```python
import json
import time
import requests

# 1. Paste the key you copied from the modal
API_KEY = "JPITYt2peEdUIVMdG9ylsuj6csIYd2MqOKs0jjid"

BASE_URL = "https://api.sportradar.com/tennis/trial/v3/en"

# The 3 endpoints required by your assignment
ENDPOINTS = {
    "competitions.json": f"{BASE_URL}/competitions.json",
    "complexes.json": f"{BASE_URL}/complexes.json",
    "double_competitor_rankings.json":
f"{BASE_URL}/double_competitor_rankings.json"
}

headers = {
    "accept": "application/json",
    "x-api-key": API_KEY
}

params = {
    "api_key": API_KEY
}

for filename, url in ENDPOINTS.items():
    print(f"Fetching {filename}...")

    # We supply the key in both query parameter and header for
compatibility
    response = requests.get(url, headers=headers, params=params)

    if response.status_code == 200:
        data = response.json()
        with open(filename, "w", encoding="utf-8") as f:
            json.dump(data, f, indent=4)
        print(f"  [SUCCESS] Saved {filename} ({len(response.content)}
bytes)")
    else:
        print(f"  [ERROR] Status: {response.status_code}")
        print(f"  Details: {response.text[:200]}")

    # Trial tier rate limit is 1 request per second
    time.sleep(1.5)

print("\nFinished downloading datasets.")
```

```
Fetching competitions.json...
  [SUCCESS] Saved competitions.json (1288218 bytes)
```

```
Fetching complexes.json...
  [SUCCESS] Saved complexes.json (749562 bytes)
Fetching double_competitor_rankings.json...
  [ERROR] Status: 404
  Details: {"generated_at":"2026-10-
06T10:02:50+00:00","message":"Invalid route."}

Finished downloading datasets.
```

```python
import os
import json
import time
import requests

# 1. Your verified API Key
API_KEY = "JPITYt2peEdUIVMdG9ylsuj6csIYd2MqOKs0jjid"

# 2. Explicit output directory on your Desktop
DESKTOP_PATH = os.path.join(os.path.expanduser("~"), "Desktop")
OUTPUT_FOLDER = os.path.join(DESKTOP_PATH, "Tennis_Data")

# Automatically create the folder if it does not exist
os.makedirs(OUTPUT_FOLDER, exist_ok=True)
print(f"Saving files directly to: {OUTPUT_FOLDER}\n")

BASE_URL = "https://api.sportradar.com/tennis/trial/v3/en"

# The 3 working endpoints identified earlier
ENDPOINTS = {
    "competitions.json": f"{BASE_URL}/competitions.json",
    "complexes.json": f"{BASE_URL}/complexes.json",
    "doubles_competitor_rankings.json": f"{BASE_URL}/rankings.json"
}

for filename, url in ENDPOINTS.items():
    target_path = os.path.join(OUTPUT_FOLDER, filename)
    print(f"Fetching {filename}...")

    response = requests.get(url, params={"api_key": API_KEY})

    if response.status_code == 200:
        with open(target_path, "w", encoding="utf-8") as f:
            json.dump(response.json(), f, indent=4)
        print(f"  -> Successfully saved: {target_path}")
    else:
        print(f"  -> Failed ({response.status_code}):
{response.text[:150]}")

    time.sleep(1.2)  # Respect API rate limits
```

```
print("\nDone! Check the 'Tennis_Data' folder sitting directly on your Desktop.")
```

```
Saving files directly to: C:\Users\prajw\Desktop\Tennis_Data

Fetching competitions.json...
  -> Successfully saved: C:\Users\prajw\Desktop\Tennis_Data\competitions.json
Fetching complexes.json...
  -> Successfully saved: C:\Users\prajw\Desktop\Tennis_Data\complexes.json
Fetching doubles_competitor_rankings.json...
  -> Successfully saved: C:\Users\prajw\Desktop\Tennis_Data\doubles_competitor_rankings.json

Done! Check the 'Tennis_Data' folder sitting directly on your Desktop.
```

SQL Code

```sql
CREATE DATABASE IF NOT EXISTS tennis_analytics;
USE tennis_analytics;

-- 1. Categories (Parent table)
CREATE TABLE IF NOT EXISTS Categories (
    category_id VARCHAR(50) PRIMARY KEY,
    category_name VARCHAR(100) NOT NULL
);

-- 2. Competitions (Child referencing Categories)
CREATE TABLE IF NOT EXISTS Competitions (
    competition_id VARCHAR(50) PRIMARY KEY,
    competition_name VARCHAR(100) NOT NULL,
    parent_id VARCHAR(50) NULL,
    type VARCHAR(50) NOT NULL,
    gender VARCHAR(20) NOT NULL,
    category_id VARCHAR(50),
    FOREIGN KEY (category_id) REFERENCES Categories(category_id) ON DELETE SET NULL
);

-- 3. Complexes (Parent table)
CREATE TABLE IF NOT EXISTS Complexes (
    complex_id VARCHAR(50) PRIMARY KEY,
    complex_name VARCHAR(100) NOT NULL
);

-- 4. Venues (Child referencing Complexes)
CREATE TABLE IF NOT EXISTS Venues (
    venue_id VARCHAR(50) PRIMARY KEY,
    venue_name VARCHAR(100) NOT NULL,
    city_name VARCHAR(100) NOT NULL,
    country_name VARCHAR(100) NOT NULL,
    country_code VARCHAR(10) NOT NULL,
    timezone VARCHAR(100) NOT NULL,
    complex_id VARCHAR(50),
    FOREIGN KEY (complex_id) REFERENCES Complexes(complex_id) ON DELETE CASCADE
);

-- 5. Competitors (Parent table)
CREATE TABLE IF NOT EXISTS Competitors (
    competitor_id VARCHAR(50) PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
    country VARCHAR(100) NOT NULL,
    country_code VARCHAR(10) NOT NULL,
    abbreviation VARCHAR(20) NOT NULL
);

-- 6. Competitor_Rankings (Child referencing Competitors)
CREATE TABLE IF NOT EXISTS Competitor_Rankings (
    rank_id INT AUTO_INCREMENT PRIMARY KEY,
```

```sql
    `rank` INT NOT NULL,
    `movement` INT NOT NULL,
    `points` INT NOT NULL,
    `competitions_played` INT NOT NULL,
    competitor_id VARCHAR(50),
    FOREIGN KEY (competitor_id) REFERENCES Competitors(competitor_id) ON DELETE CASCADE
);
```

```python
import json
import os
import pandas as pd
from sqlalchemy import create_engine, text
from sqlalchemy.engine import URL

# --- 1. Database Connection Setup ---
# URL.create passes your custom password directly without parsing issues
connection_url = URL.create(
    drivername="mysql+pymysql",
    username="root",
    password="PradiAtMysql24",  # <-- Replace with your new custom password
    host="127.0.0.1",
    port=3306,
    database="tennis_analytics"
)

engine = create_engine(connection_url)

# --- 2. Test Connection ---
def test_connection():
    print("Connecting to MySQL database...")
    try:
        with engine.connect() as conn:
            result = conn.execute(text("SELECT DATABASE();"))
            db_name = result.scalar()
            print(f"[SUCCESS] Connected to database: {db_name}\n")
            return True
    except Exception as e:
        print(f"[ERROR] Connection failed: {e}")
        return False

# --- 3. Ingestion Functions ---
def load_competitions_data():
    if not os.path.exists("competitions.json"):
        print("[SKIP] competitions.json not found.")
        return

    print("Loading Competitions & Categories...")
    with open("competitions.json", "r", encoding="utf-8") as f:
        data = json.load(f)

    categories = []
    competitions = []

    for comp in data.get("competitions", []):
        cat = comp.get("category", {})
        cat_id = cat.get("id") if cat else None

        if cat_id:
            categories.append({
                "category_id": cat_id,
                "category_name": cat.get("name", "Unknown")
            })

        competitions.append({
            "competition_id": comp.get("id"),
            "competition_name": comp.get("name", "Unknown"),
            "parent_id": comp.get("parent_id"),
            "type": comp.get("type", "unknown"),
            "gender": comp.get("gender", "unknown"),
            "category_id": cat_id
        })

    df_cat = pd.DataFrame(categories).drop_duplicates(subset=["category_id"])
    df_comp = pd.DataFrame(competitions).drop_duplicates(subset=["competition_id"])

    df_cat.to_sql("Categories", con=engine, if_exists="append", index=False)
    df_comp.to_sql("Competitions", con=engine, if_exists="append", index=False)
    print(f" -> Inserted {len(df_cat)} categories and {len(df_comp)} competitions.")

def load_complexes_data():
```

```
    if not os.path.exists("complexes.json"):
        print("[SKIP] complexes.json not found.")
        return

    print("Loading Complexes & Venues...")
    with open("complexes.json", "r", encoding="utf-8") as f:
        data = json.load(f)

    complexes = []
    venues = []

    for comp in data.get("complexes", []):
        c_id = comp.get("id")
        complexes.append({
            "complex_id": c_id,
            "complex_name": comp.get("name", "Unknown")
        })

        for v in comp.get("venues", []):
            venues.append({
                "venue_id": v.get("id"),
                "venue_name": v.get("name", "Unknown"),
                "city_name": v.get("city_name", "Unknown"),
                "country_name": v.get("country_name", "Unknown"),
                "country_code": (v.get("country_code") or "UNK")[:10],
                "timezone": v.get("timezone", "UTC"),
                "complex_id": c_id
            })

    df_complexes = pd.DataFrame(complexes).drop_duplicates(subset=["complex_id"])
    df_venues = pd.DataFrame(venues).drop_duplicates(subset=["venue_id"])

    df_complexes.to_sql("Complexes", con=engine, if_exists="append", index=False)
    df_venues.to_sql("Venues", con=engine, if_exists="append", index=False)
    print(f" -> Inserted {len(df_complexes)} complexes and {len(df_venues)} venues.")

def load_rankings_data():
    filename = "doubles_competitor_rankings.json"
    if not os.path.exists(filename):
        print(f"[SKIP] {filename} not found.")
        return

    print("Loading Competitors & Rankings...")
    with open(filename, "r", encoding="utf-8") as f:
        data = json.load(f)

    competitors = []
    rankings = []

    for group in data.get("rankings", []):
        for item in group.get("competitor_rankings", []):
            comp = item.get("competitor", {})
            c_id = comp.get("id")
            if not c_id:
                continue

            competitors.append({
                "competitor_id": c_id,
                "name": comp.get("name", "Unknown"),
                "country": comp.get("country", "Unknown"),
                "country_code": (comp.get("country_code") or "UNK")[:10],
                "abbreviation": (comp.get("abbreviation") or "UNK")[:20]
            })

            rankings.append({
                "rank": item.get("rank", 0),
                "movement": item.get("movement", 0),
                "points": item.get("points", 0),
                "competitions_played": item.get("competitions_played", 0),
                "competitor_id": c_id
            })
```

```
    df_competitors = pd.DataFrame(competitors).drop_duplicates(subset=["competitor_id"])
    df_rankings = pd.DataFrame(rankings)

    df_competitors.to_sql("Competitors", con=engine, if_exists="append", index=False)
    df_rankings.to_sql("Competitor_Rankings", con=engine, if_exists="append", index=False)
    print(f" -> Inserted {len(df_competitors)} competitors and {len(df_rankings)} rankings.")

if __name__ == "__main__":
    if test_connection():
        load_competitions_data()
        load_complexes_data()
        load_rankings_data()
        print("\nAll database tables populated successfully!")
```

```sql
USE tennis_analytics;

-- ========================================================================
-- 1. COMPETITIONS ANALYSIS
-- ========================================================================

-- 1.1 List all competitions along with their category name
SELECT
    c.competition_id,
    c.competition_name,
    cat.category_name,
    c.type,
    c.gender
FROM Competitions c
LEFT JOIN Categories cat ON c.category_id = cat.category_id;
```

Output is in next page.

```
13          c.gender
14      FROM Competitions c
15      LEFT JOIN Categories cat ON c.category_id = cat.category_id;
16
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

competition_id	competition_name	category_name	type	gender
sr:competition:10025	ITF Men Stara Zagora, Bulgaria Men Singles	ITF Men	singles	men
sr:competition:10027	ITF Men Stara Zagora, Bulgaria Men Doubles	ITF Men	doubles	men
sr:competition:10031	ITF Men Sibiu, Romania Men Singles	ITF Men	singles	men
sr:competition:10033	ITF Men Sibiu, Romania Men Doubles	ITF Men	doubles	men
sr:competition:10037	ITF Men Busto Arsizio, Italy Men Singles	ITF Men	singles	men
sr:competition:10039	ITF Men Busto Arsizio, Italy Men Doubles	ITF Men	doubles	men
sr:competition:10043	ITF Men Sabac, Serbia Men Singles	ITF Men	singles	men
sr:competition:10045	ITF Men Sabac, Serbia Men Doubles	ITF Men	doubles	men
sr:competition:10055	ITF Men Seefeld, Austria Men Singles	ITF Men	singles	men

singles

Result 5

Output

Action Output

#	Time	Action	Message
1	17:27:54	USE tennis_analytics	0 row(s) affected
2	17:28:21	SELECT c.competition_id, c.competition_name, cat.category_name, c.type, ...	6694 row(s) returned

```sql
-- 1.2 Count the number of competitions in each category
SELECT
    cat.category_name,
    COUNT(c.competition_id) AS competition_count
FROM Categories cat
LEFT JOIN Competitions c ON cat.category_id = c.category_id
GROUP BY cat.category_name
ORDER BY competition_count DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

category_name	competition_count
ITF Men	2198
ITF Women	2032
Challenger	1065
UTR Men	307
WTA	259
WTA 125K	258
UTR Women	256
ATP	225
Exhibition	38

Result 6

Output

Action Output

#	Time	Action	Message
1	17:27:54	USE tennis_analytics	0 row(s) affected
2	17:28:21	SELECT c.competition_id, c.competition_name, cat.category_name, c.type, ...	6694 row(s) returned
3	17:29:06	SELECT cat.category_name, COUNT(c.competition_id) AS competition_count FR...	18 row(s) returned

```sql
-- 1.3 Find all competitions of type 'doubles'
SELECT
    competition_id,
    competition_name,
    type,
    gender
FROM Competitions
WHERE LOWER(type) = 'doubles';
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: | Fetch rows:

competition_id	competition_name	type	gender
sr:competition:10027	ITF Men Stara Zagora, Bulgaria Men Doubles	doubles	men
sr:competition:10033	ITF Men Sibiu, Romania Men Doubles	doubles	men
sr:competition:10039	ITF Men Busto Arsizio, Italy Men Doubles	doubles	men
sr:competition:10045	ITF Men Sabac, Serbia Men Doubles	doubles	men
sr:competition:10057	ITF Men Seefeld, Austria Men Doubles	doubles	men
sr:competition:10075	ATP Challenger Portoroz, Slovenia Men Doubles...	doubles	men
sr:competition:10093	ITF Men Focsani, Romania Men Doubles	doubles	men
sr:competition:10153	ATP Challenger Eskisehir, Turkey Men Doubles	doubles	men
sr:competition:10159	ITF Men Tallinn, Estonia Men Doubles	doubles	men

Competitions 7

Output

Action Output

#	Time	Action	Message
1	17:27:54	USE tennis_analytics	0 row(s) affected
2	17:28:21	SELECT c.competition_id, c.competition_name, cat.category_name, c.type, ...	6694 row(s) returned
3	17:29:06	SELECT cat.category_name, COUNT(c.competition_id) AS competition_count FR...	18 row(s) returned
4	17:30:56	SELECT competition_id, competition_name, type, gender FROM Competitions...	3051 row(s) returned

```sql
-- 1.4 Get competitions that belong to a specific category (e.g., ITF Men)
SELECT
    c.competition_id,
    c.competition_name,
    cat.category_name
FROM Competitions c
JOIN Categories cat ON c.category_id = cat.category_id
WHERE cat.category_name = 'ITF Men';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

competition_id	competition_name	category_name
sr:competition:10025	ITF Men Stara Zagora, Bulgaria Men Singles	ITF Men
sr:competition:10027	ITF Men Stara Zagora, Bulgaria Men Doubles	ITF Men
sr:competition:10031	ITF Men Sibiu, Romania Men Singles	ITF Men
sr:competition:10033	ITF Men Sibiu, Romania Men Doubles	ITF Men
sr:competition:10037	ITF Men Busto Arsizio, Italy Men Singles	ITF Men
sr:competition:10039	ITF Men Busto Arsizio, Italy Men Doubles	ITF Men
sr:competition:10043	ITF Men Sabac, Serbia Men Singles	ITF Men
sr:competition:10045	ITF Men Sabac, Serbia Men Doubles	ITF Men
sr:competition:10055	ITF Men Seefeld, Austria Men Singles	ITF Men

Export recordset to an external file

Result 8

Output

Action Output

#	Time	Action	Message
1	17:27:54	USE tennis_analytics	0 row(s) affected
2	17:28:21	SELECT c.competition_id, c.competition_name, cat.category_name, c.type, ...	6694 row(s) returned
3	17:29:06	SELECT cat.category_name, COUNT(c.competition_id) AS competition_count FR...	18 row(s) returned
4	17:30:56	SELECT competition_id, competition_name, type, gender FROM Competitions...	3051 row(s) returned
5	17:31:53	SELECT c.competition_id, c.competition_name, cat.category_name FROM Com...	2198 row(s) returned

```sql
-- 1.5 Identify parent competitions and their sub-competitions
SELECT
    parent.competition_name AS parent_competition,
    child.competition_name AS sub_competition
FROM Competitions child
JOIN Competitions parent ON child.parent_id = parent.competition_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

parent_competition	sub_competition
ITF Romania F9, Men Singles	ITF Romania F9, Men Doubles

Result 9

Output

Action Output

#	Time	Action	Message
1	17:27:54	USE tennis_analytics	0 row(s) affected
2	17:28:21	SELECT c.competition_id, c.competition_name, cat.category_name, c.type, ...	6694 row(s) returned
3	17:29:06	SELECT cat.category_name, COUNT(c.competition_id) AS competition_count FR...	18 row(s) returned
4	17:30:56	SELECT competition_id, competition_name, type, gender FROM Competitions...	3051 row(s) returned
5	17:31:53	SELECT c.competition_id, c.competition_name, cat.category_name FROM Com...	2198 row(s) returned
6	17:34:41	SELECT parent.competition_name AS parent_competition, child.competition_name...	1 row(s) returned

```sql
-- 1.6 Analyze the distribution of competition types by category
SELECT
    cat.category_name,
    c.type,
    COUNT(c.competition_id) AS total_competitions
FROM Categories cat
JOIN Competitions c ON cat.category_id = c.category_id
GROUP BY cat.category_name, c.type
ORDER BY cat.category_name, total_competitions DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

category_name	type	total_competitions
ATP	singles	109
ATP	doubles	108
ATP	mixed_doubles	6
ATP	mixed	2
Billie Jean King Cup	mixed	1
Challenger	singles	533
Challenger	doubles	532
Davis Cup	mixed	1
Exhibition	singles	27

Result 10

Output

Action Output

#	Time	Action	Message
2	17:28:21	SELECT c.competition_id, c.competition_name, cat.category_name, c.type,...	6694 row(s) returned
3	17:29:06	SELECT cat.category_name, COUNT(c.competition_id) AS competition_count F...	18 row(s) returned
4	17:30:56	SELECT competition_id, competition_name, type, gender FROM Competitio...	3051 row(s) returned
5	17:31:53	SELECT c.competition_id, c.competition_name, cat.category_name FROM Co...	2198 row(s) returned
6	17:34:41	SELECT parent.competition_name AS parent_competition, child.competition_na...	1 row(s) returned
7	17:35:40	SELECT cat.category_name, c.type, COUNT(c.competition_id) AS total_comp...	32 row(s) returned

```sql
-- 1.7 List all competitions with no parent (top-level competitions)
SELECT
    competition_id,
    competition_name,
    type,
    gender
FROM Competitions
WHERE parent_id IS NULL;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

competition_id	competition_name	type	gender
sr:competition:1207	Championship International Series	singles	women
sr:competition:12229	IPTL	singles	men
sr:competition:15393	ITF Men San Jose, Costa		
sr:competition:15395	ITF Men San Jose, Costa Rica Men Doubles	doubles	men
sr:competition:2100	Davis Cup	mixed	men
sr:competition:2102	Billie Jean King Cup	mixed	women
sr:competition:21028	ITF Colombia 02A, Women Singles	singles	women
sr:competition:21228	ITF Australia 02B, Women Doubles	doubles	women
sr:competition:22334	ITF Uzbekistan 01A, Women Singles	singles	women

ITF Men San Jose, Costa Rica Men Singles

Competitions 11 ×

Output

Action Output

#	Time	Action	Message
3	17:29:06	SELECT cat.category_name, COUNT(c.competition_id) AS competition_count F...	18 row(s) returned
4	17:30:56	SELECT competition_id, competition_name, type, gender FROM Competitio...	3051 row(s) returned
5	17:31:53	SELECT c.competition_id, c.competition_name, cat.category_name FROM Co...	2198 row(s) returned
6	17:34:41	SELECT parent.competition_name AS parent_competition, child.competition_na...	1 row(s) returned
7	17:35:40	SELECT cat.category_name, c.type, COUNT(c.competition_id) AS total_comp...	32 row(s) returned
8	17:37:42	SELECT competition_id, competition_name, type, gender FROM Competitio...	608 row(s) returned

```sql
-- ========================================================================
-- 2. VENUE ANALYSIS
-- ========================================================================

-- 2.1 List all venues along with their associated complex name
SELECT
    v.venue_name,
    v.city_name,
    v.country_name,
    c.complex_name
FROM Venues v
JOIN Complexes c ON v.complex_id = c.complex_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

venue_name	city_name	country_name	complex_name
Estadio Monumental	Lima	Peru	Club Tennis Las Terrazas de Miraflores
Stadium	Shenzhen	China	Shenzhen International Tennis Center
Court 2	Shenzhen	China	Shenzhen Longgang Sports Center
Court 7	Sydney	Australia	Olympic Park Tennis Center
Court 20	Melbourne	Australia	Melbourne Park
Court 22	Melbourne	Australia	Melbourne Park
Estadio Ivan Elias Moreno	Lima	Peru	Club Tennis Las Terrazas de Miraflores
Stade Numa-Daly Magenta	Noumea	New Caledonia	Complexe Tennistique Marie-Louise Lhuillier
Cancha 5	Acapulco	Mexico	Acapulco Princess Mundo Imperial

Melbourne Park

Result 12

Output

Action Output

#	Time	Action	Message
4	17:30:56	SELECT competition_id, competition_name, type, gender FROM Competitio...	3051 row(s) returned
5	17:31:53	SELECT c.competition_id, c.competition_name, cat.category_name FROM Co...	2198 row(s) returned
6	17:34:41	SELECT parent.competition_name AS parent_competition, child.competition_na...	1 row(s) returned
7	17:35:40	SELECT cat.category_name, c.type, COUNT(c.competition_id) AS total_comp...	32 row(s) returned
8	17:37:42	SELECT competition_id, competition_name, type, gender FROM Competitio...	608 row(s) returned
9	17:38:45	SELECT v.venue_name, v.city_name, v.country_name, c.complex_name F...	4124 row(s) returned

```sql
-- 2.2 Count the number of venues in each complex
SELECT
    c.complex_name,
    COUNT(v.venue_id) AS venue_count
FROM Complexes c
LEFT JOIN Venues v ON c.complex_id = v.complex_id
GROUP BY c.complex_name
ORDER BY venue_count DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

complex_name	venue_count
National Tennis Center	75
Buenos Aires Lawn Tennis Club	29
Melbourne Park	28
Queensland Tennis Centre	25
Hurd Tennis Center	24
Foro Italico	23
AELTC Wimbledon Qualifying and Community Sp...	23
Club Tennis Las Terrazas de Miraflores	22
Indian Wells Tennis Garden	21

Result 13

Output

Action Output

#	Time	Action	Message
5	17:31:53	SELECT c.competition_id, c.competition_name, cat.category_name FROM Co...	2198 row(s) returned
6	17:34:41	SELECT parent.competition_name AS parent_competition, child.competition_na...	1 row(s) returned
7	17:35:40	SELECT cat.category_name, c.type, COUNT(c.competition_id) AS total_comp...	32 row(s) returned
8	17:37:42	SELECT competition_id, competition_name, type, gender FROM Competitio...	608 row(s) returned
9	17:38:45	SELECT v.venue_name, v.city_name, v.country_name, c.complex_name F...	4124 row(s) returned
10	17:40:25	SELECT c.complex_name, COUNT(v.venue_id) AS venue_count FROM Comple...	756 row(s) returned

```sql
-- 2.3 Get details of venues in a specific country (e.g., Chile or Great Britain)
SELECT
    venue_name,
    city_name,
    country_name,
    country_code,
    timezone
FROM Venues
WHERE country_name = 'Chile';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

venue_name	city_name	country_name	country_code	timezone
Cancha 1	Santiago	Chile	CHL	America/Santiago
Cancha 3	Santiago	Chile	CHL	America/Santiago
Centre Court	Santiago	Chile	CHL	America/Santiago
Court 16	Santiago	Chile	CHL	America/Santiago
Court 15	Santiago	Chile	CHL	America/Santiago
Court 11	Santiago	Chile	CHL	America/Santiago
Cancha Central	Santiago	Chile	CHL	America/Santiago
Court 2	Santiago	Chile	CHL	America/Santiago
Cancha 4	Santiago	Chile	CHL	America/Santiago

Venues 14

Output

Action Output

#	Time	Action	Message
6	17:34:41	SELECT parent.competition_name AS parent_competition, child.competition_na...	1 row(s) returned
7	17:35:40	SELECT cat.category_name, c.type, COUNT(c.competition_id) AS total_comp...	32 row(s) returned
8	17:37:42	SELECT competition_id, competition_name, type, gender FROM Competitio...	608 row(s) returned
9	17:38:45	SELECT v.venue_name, v.city_name, v.country_name, c.complex_name F...	4124 row(s) returned
10	17:40:25	SELECT c.complex_name, COUNT(v.venue_id) AS venue_count FROM Comple...	756 row(s) returned
11	17:41:16	SELECT venue_name, city_name, country_name, country_code, timezon...	65 row(s) returned

```sql
-- 2.4 Identify all venues and their timezones
SELECT
    venue_name,
    country_name,
    timezone
FROM Venues
ORDER BY timezone;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

venue_name	country_name	timezone
Central KAYDAN	Cote d'Ivoire	Africa/Abidjan
Court CNPS	Cote d'Ivoire	Africa/Abidjan
Court 1 COFINA NEEMBA	Cote d'Ivoire	Africa/Abidjan
Court 2 NYCKSOIL	Cote d'Ivoire	Africa/Abidjan
Court 4	Cote d'Ivoire	Africa/Abidjan
Centre Court	Egypt	Africa/Cairo
Court 3	Egypt	Africa/Cairo
Court 2	Egypt	Africa/Cairo
Court 10	Egypt	Africa/Cairo

Venues 15

Output

Action Output

#	Time	Action	Message
7	17:35:40	SELECT cat.category_name, c.type, COUNT(c.competition_id) AS total_comp...	32 row(s) returned
8	17:37:42	SELECT competition_id, competition_name, type, gender FROM Competitio...	608 row(s) returned
9	17:38:45	SELECT v.venue_name, v.city_name, v.country_name, c.complex_name F...	4124 row(s) returned
10	17:40:25	SELECT c.complex_name, COUNT(v.venue_id) AS venue_count FROM Comple...	756 row(s) returned
11	17:41:16	SELECT venue_name, city_name, country_name, country_code, timezon...	65 row(s) returned
12	17:42:18	SELECT venue_name, country_name, timezone FROM Venues ORDER BY ti...	4124 row(s) returned

```sql
-- 2.5 Find complexes that have more than one venue
SELECT
    c.complex_name,
    COUNT(v.venue_id) AS venue_count
FROM Complexes c
JOIN Venues v ON c.complex_id = v.complex_id
GROUP BY c.complex_name
HAVING COUNT(v.venue_id) > 1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

complex_name	venue_count
Kindarena	4
Estadio de La Cartuja	2
Estadio German Becker	4
Palais des sports de Gerland	7
Sibur Arena	4
Palais Des Sport	7
Palasport	4
Salle Steredenn	7
Porsche-Arena	3

Result 16

Output

Action Output

#	Time	Action	Message
8	17:37:42	SELECT competition_id, competition_name, type, gender FROM Competitio...	608 row(s) returned
9	17:38:45	SELECT v.venue_name, v.city_name, v.country_name, c.complex_name F...	4124 row(s) returned
10	17:40:25	SELECT c.complex_name, COUNT(v.venue_id) AS venue_count FROM Comple...	756 row(s) returned
11	17:41:16	SELECT venue_name, city_name, country_name, country_code, timezon...	65 row(s) returned
12	17:42:18	SELECT venue_name, country_name, timezone FROM Venues ORDER BY ti...	4124 row(s) returned
13	17:43:15	SELECT c.complex_name, COUNT(v.venue_id) AS venue_count FROM Comple...	583 row(s) returned

```sql
-- 2.6 List venues grouped by country
SELECT
    country_name,
    COUNT(venue_id) AS total_venues
FROM Venues
GROUP BY country_name
ORDER BY total_venues DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

country_name	total_venues
USA	677
Italy	300
France	276
China	241
Spain	217
Germany	182
Australia	144
Mexico	144
England	117

Result 17

Output

Action Output

#	Time	Action	Message
9	17:38:45	SELECT v.venue_name, v.city_name, v.country_name, c.complex_name F...	4124 row(s) returned
10	17:40:25	SELECT c.complex_name, COUNT(v.venue_id) AS venue_count FROM Comple...	756 row(s) returned
11	17:41:16	SELECT venue_name, city_name, country_name, country_code, timezon...	65 row(s) returned
12	17:42:18	SELECT venue_name, country_name, timezone FROM Venues ORDER BY ti...	4124 row(s) returned
13	17:43:15	SELECT c.complex_name, COUNT(v.venue_id) AS venue_count FROM Comple...	583 row(s) returned
14	17:44:13	SELECT country_name, COUNT(venue_id) AS total_venues FROM Venues GR...	77 row(s) returned

```sql
-- 2.7 Find all venues for a specific complex (e.g., Nacional)
SELECT
    v.venue_name,
    v.city_name,
    v.country_name,
    c.complex_name
FROM Venues v
JOIN Complexes c ON v.complex_id = c.complex_id
WHERE c.complex_name LIKE '%Nacional%';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

venue_name	city_name	country_name	complex_name
Estadio CIF	Lisbon	Portugal	Club Internacional de Foot-ball
Court Lisboa	Lisbon	Portugal	Club Internacional de Foot-ball
Court Belem	Lisbon	Portugal	Club Internacional de Foot-ball
Court 5	Lisbon	Portugal	Club Internacional de Foot-ball
Court 7	Lisbon	Portugal	Club Internacional de Foot-ball
Court Central	Lisbon	Portugal	Club Internacional de Foot-ball
Court 9	Lisbon	Portugal	Club Internacional de Foot-ball
Court 8	Lisbon	Portugal	Club Internacional de Foot-ball
Court 6	Lisbon	Portugal	Club Internacional de Foot-ball

Result 18

Output

Action Output

#	Time	Action	Message
10	17:40:25	SELECT c.complex_name, COUNT(v.venue_id) AS venue_count FROM Comple...	756 row(s) returned
11	17:41:16	SELECT venue_name, city_name, country_name, country_code, timezon...	65 row(s) returned
12	17:42:18	SELECT venue_name, country_name, timezone FROM Venues ORDER BY ti...	4124 row(s) returned
13	17:43:15	SELECT c.complex_name, COUNT(v.venue_id) AS venue_count FROM Comple...	583 row(s) returned
14	17:44:13	SELECT country_name, COUNT(venue_id) AS total_venues FROM Venues GR...	77 row(s) returned
15	17:45:43	SELECT v.venue_name, v.city_name, v.country_name, c.complex_name F...	26 row(s) returned

```sql
-- ========================================================================
-- 3. COMPETITOR RANKING ANALYSIS
-- ========================================================================

-- 3.1 Get all competitors with their rank and points
SELECT
    c.name,
    c.country,
    r.`rank`,
    r.points
FROM Competitor_Rankings r
JOIN Competitors c ON r.competitor_id = c.competitor_id
ORDER BY r.`rank` ASC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

name	country	rank	points
Sinner, Jannik	Italy	1	11000
Rybakina, Elena	Kazakhstan	1	9901
Zverev, Alexander	Germany	2	9720
Sabalenka, Aryna	Neutral	2	7810
Alcaraz, Carlos	Spain	3	5260
Pegula, Jessica	USA	3	7265
Shelton, Ben	USA	4	4680
Gauff, Coco	USA	4	7244
Medvedev, Daniil	Neutral	5	4010

Result 19

Output

Action Output

#	Time	Action	Message
11	17:41:16	SELECT venue_name, city_name, country_name, country_code, timezon...	65 row(s) returned
12	17:42:18	SELECT venue_name, country_name, timezone FROM Venues ORDER BY ti...	4124 row(s) returned
13	17:43:15	SELECT c.complex_name, COUNT(v.venue_id) AS venue_count FROM Comple...	583 row(s) returned
14	17:44:13	SELECT country_name, COUNT(venue_id) AS total_venues FROM Venues GR...	77 row(s) returned
15	17:45:43	SELECT v.venue_name, v.city_name, v.country_name, c.complex_name F...	26 row(s) returned
16	17:46:47	SELECT c.name, c.country, r.`rank`, r.points FROM Competitor_Rankings r ...	1000 row(s) returned

```sql
-- 3.2 Find competitors ranked in the top 5
SELECT
    c.name,
    c.country,
    r.`rank`,
    r.points
FROM Competitor_Rankings r
JOIN Competitors c ON r.competitor_id = c.competitor_id
ORDER BY r.`rank` ASC
LIMIT 5;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

name	country	rank	points
Sinner, Jannik	Italy	1	11000
Rybakina, Elena	Kazakhstan	1	9901
Zverev, Alexander	Germany	2	9720
Sabalenka, Aryna	Neutral	2	7810
Alcaraz, Carlos	Spain	3	5260

Result 20

Output

Action Output

#	Time	Action	Message
12	17:42:18	SELECT venue_name, country_name, timezone FROM Venues ORDER BY ti...	4124 row(s) returned
13	17:43:15	SELECT c.complex_name, COUNT(v.venue_id) AS venue_count FROM Comple...	583 row(s) returned
14	17:44:13	SELECT country_name, COUNT(venue_id) AS total_venues FROM Venues GR...	77 row(s) returned
15	17:45:43	SELECT v.venue_name, v.city_name, v.country_name, c.complex_name F...	26 row(s) returned
16	17:46:47	SELECT c.name, c.country, r.`rank`, r.points FROM Competitor_Rankings r ...	1000 row(s) returned
17	17:48:27	SELECT c.name, c.country, r.`rank`, r.points FROM Competitor_Rankings r ...	5 row(s) returned

```sql
-- 3.3 List competitors with no rank movement (stable rank)
SELECT
    c.name,
    c.country,
    r.`rank`,
    r.movement
FROM Competitor_Rankings r
JOIN Competitors c ON r.competitor_id = c.competitor_id
WHERE r.movement = 0
ORDER BY r.`rank` ASC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

name	country	rank	movement
Sinner, Jannik	Italy	1	0
Rybakina, Elena	Kazakhstan	1	0
Zverev, Alexander	Germany	2	0
Sabalenka, Aryna	Neutral	2	0
Alcaraz, Carlos	Spain	3	0
Pegula, Jessica	USA	3	0
Shelton, Ben	USA	4	0
Gauff, Coco	USA	4	0
Andreeva, Mirra	Neutral	5	0

Result 21

Output

Action Output

#	Time	Action	Message
13	17:43:15	SELECT c.complex_name, COUNT(v.venue_id) AS venue_count FROM Comple...	583 row(s) returned
14	17:44:13	SELECT country_name, COUNT(venue_id) AS total_venues FROM Venues GR...	77 row(s) returned
15	17:45:43	SELECT v.venue_name, v.city_name, v.country_name, c.complex_name F...	26 row(s) returned
16	17:46:47	SELECT c.name, c.country, r.`rank`, r.points FROM Competitor_Rankings r ...	1000 row(s) returned
17	17:48:27	SELECT c.name, c.country, r.`rank`, r.points FROM Competitor_Rankings r ...	5 row(s) returned
18	17:49:37	SELECT c.name, c.country, r.`rank`, r.movement FROM Competitor_Rankin...	90 row(s) returned

```sql
-- 3.4 Get the total points of competitors from a specific country (e.g., Croatia)
SELECT
    c.country,
    SUM(r.points) AS total_points,
    COUNT(c.competitor_id) AS total_players
FROM Competitors c
JOIN Competitor_Rankings r ON c.competitor_id = r.competitor_id
WHERE c.country = 'Croatia'
GROUP BY c.country;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

country	total_points	total_players
Croatia	6557	14

Result 22

Output

Action Output

#	Time	Action	Message
14	17:44:13	SELECT country_name, COUNT(venue_id) AS total_venues FROM Venues GR...	77 row(s) returned
15	17:45:43	SELECT v.venue_name, v.city_name, v.country_name, c.complex_name F...	26 row(s) returned
16	17:46:47	SELECT c.name, c.country, r.`rank`, r.points FROM Competitor_Rankings r ...	1000 row(s) returned
17	17:48:27	SELECT c.name, c.country, r.`rank`, r.points FROM Competitor_Rankings r ...	5 row(s) returned
18	17:49:37	SELECT c.name, c.country, r.`rank`, r.movement FROM Competitor_Rankin...	90 row(s) returned
19	17:50:38	SELECT c.country, SUM(r.points) AS total_points, COUNT(c.competitor_id) AS...	1 row(s) returned

```sql
-- 3.5 Count the number of competitors per country
SELECT
    country,
    COUNT(competitor_id) AS competitor_count
FROM Competitors
GROUP BY country
ORDER BY competitor_count DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

country	competitor_count
USA	100
France	77
Italy	66
Neutral	54
Spain	48
Great Britain	46
Japan	42
Argentina	40
Germany	40

Result 23

Output

Action Output

#	Time	Action	Message
16	17:46:47	SELECT c.name, c.country, r.`rank`, r.points FROM Competitor_Rankings r ...	1000 row(s) returned
17	17:48:27	SELECT c.name, c.country, r.`rank`, r.points FROM Competitor_Rankings r ...	5 row(s) returned
18	17:49:37	SELECT c.name, c.country, r.`rank`, r.movement FROM Competitor_Rankin...	90 row(s) returned
19	17:50:38	SELECT c.country, SUM(r.points) AS total_points, COUNT(c.competitor_id) AS...	1 row(s) returned
20	17:51:40	SELECT country, COUNT(competitor_id) AS competitor_count FROM Competitor...	Error Code: 1064. You have
21	17:51:49	SELECT country, COUNT(competitor_id) AS competitor_count FROM Competitor...	79 row(s) returned

```sql
-- 3.6 Find competitors with the highest points in the current week
SELECT
    c.name,
    c.country,
    r.`rank`,
    r.points
FROM Competitor_Rankings r
JOIN Competitors c ON r.competitor_id = c.competitor_id
ORDER BY r.points DESC
LIMIT 10;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

name	country	rank	points
Sinner, Jannik	Italy	1	11000
Rybakina, Elena	Kazakhstan	1	9901
Zverev, Alexander	Germany	2	9720
Sabalenka, Aryna	Neutral	2	7810
Pegula, Jessica	USA	3	7265
Gauff, Coco	USA	4	7244
Andreeva, Mirra	Neutral	5	5841
Noskova, Linda	Czechia	6	5328
Alcaraz, Carlos	Spain	3	5260

esult 24 ×

Output

Action Output

#	Time	Action	Message
17	17:48:27	SELECT c.name, c.country, r.`rank`, r.points FROM Competitor_Rankings r ...	5 row(s) returned
18	17:49:37	SELECT c.name, c.country, r.`rank`, r.movement FROM Competitor_Rankin...	90 row(s) returned
19	17:50:38	SELECT c.country, SUM(r.points) AS total_points, COUNT(c.competitor_id) AS...	1 row(s) returned
20	17:51:40	SELECT country, COUNT(competitor_id) AS competitor_count FROM Competitor...	Error Code: 1064. You h
21	17:51:49	SELECT country, COUNT(competitor_id) AS competitor_count FROM Competitor...	79 row(s) returned
22	17:52:50	SELECT c.name, c.country, r.`rank`, r.points FROM Competitor_Rankings r ...	10 row(s) returned

```python
import streamlit as st
import pandas as pd
from sqlalchemy import create_engine, text
from sqlalchemy.engine import URL
import plotly.express as px

# ------------------------------------------------------------
# Page Configuration
# ------------------------------------------------------------
st.set_page_config(
    page_title="Tennis Global Analytics",
    layout="wide",
    initial_sidebar_state="expanded"
)

# ------------------------------------------------------------
# Database Connection
# ------------------------------------------------------------
DB_USER = "root"
DB_PASS = "PradiAtMysql24"  # <-- Enter your MySQL password here
DB_HOST = "127.0.0.1"
DB_PORT = 3306
DB_NAME = "tennis_analytics"

@st.cache_resource
def get_database_engine():
    connection_url = URL.create(
        drivername="mysql+pymysql",
        username=DB_USER,
        password=DB_PASS,
        host=DB_HOST,
        port=DB_PORT,
        database=DB_NAME
    )
    return create_engine(connection_url)

engine = get_database_engine()

@st.cache_data(ttl=600)
def run_query(sql_query, params=None):
    with engine.connect() as conn:
        return pd.read_sql(text(sql_query), conn, params=params)

# ------------------------------------------------------------
# Sidebar Navigation
# ------------------------------------------------------------
st.sidebar.title("Tennis Analytics")
menu = st.sidebar.radio(
    "Navigate Modules",
    ["Executive Overview", "Competitions Explorer", "Venue Infrastructure", "Competitor
Rankings", "SQL Query Sandbox"]
)

# ------------------------------------------------------------
# 1. Executive Overview
# ------------------------------------------------------------
if menu == "Executive Overview":
    st.title(" Tennis Ecosystem Global Overview")
    st.markdown("High-level KPI distribution across professional circuits, global venues, and
active competitors.")

    # High-level KPIs
    col1, col2, col3, col4 = st.columns(4)
    total_comps = run_query("SELECT COUNT(*) AS val FROM Competitions")['val'][0]
    total_venues = run_query("SELECT COUNT(*) AS val FROM Venues")['val'][0]
    total_complexes = run_query("SELECT COUNT(*) AS val FROM Complexes")['val'][0]
    total_players = run_query("SELECT COUNT(*) AS val FROM Competitors")['val'][0]

    col1.metric("Total Competitions", f"{total_comps:,}")
    col2.metric("Total Venues", f"{total_venues:,}")
    col3.metric("Tennis Complexes", f"{total_complexes:,}")
    col4.metric("Ranked Competitors", f"{total_players:,}")
```

```
    st.markdown("---")

    col_left, col_right = st.columns(2)

    with col_left:
        st.subheader("Tournament Volume by Category")
        df_cat = run_query("""
            SELECT cat.category_name, COUNT(c.competition_id) AS total
            FROM Categories cat
            JOIN Competitions c ON cat.category_id = c.category_id
            GROUP BY cat.category_name
            ORDER BY total DESC
            LIMIT 10;
        """)
        fig_cat = px.bar(df_cat, x="total", y="category_name", orientation="h",
                         color="total", color_continuous_scale="Viridis",
                         labels={"total": "Competitions", "category_name": "Circuit Category"})
        fig_cat.update_layout(yaxis={'categoryorder': 'total ascending'}, showlegend=False)
        st.plotly_chart(fig_cat, use_container_width=True)

    with col_right:
        st.subheader("Top 10 Host Countries by Venues")
        df_venues_ctry = run_query("""
            SELECT country_name, COUNT(venue_id) AS total_venues
            FROM Venues
            GROUP BY country_name
            ORDER BY total_venues DESC
            LIMIT 10;
        """)
        fig_venue = px.pie(df_venues_ctry, values="total_venues", names="country_name",
                           hole=0.4, color_discrete_sequence=px.colors.sequential.Teal)
        st.plotly_chart(fig_venue, use_container_width=True)

# ------------------------------------------------------------
# 2. Competitions Explorer
# ------------------------------------------------------------
elif menu == "Competitions Explorer":
    st.title("Competitions & Circuits Analysis")

    cat_list = run_query("SELECT DISTINCT category_name FROM Categories ORDER BY
category_name;")['category_name'].tolist()
    selected_cat = st.selectbox("Filter by Category", ["All"] + cat_list)

    if selected_cat == "All":
        query_comp = """
            SELECT c.competition_id, c.competition_name, cat.category_name, c.type, c.gender
            FROM Competitions c
            LEFT JOIN Categories cat ON c.category_id = cat.category_id;
        """
        df_comp = run_query(query_comp)
    else:
        query_comp = """
            SELECT c.competition_id, c.competition_name, cat.category_name, c.type, c.gender
            FROM Competitions c
            JOIN Categories cat ON c.category_id = cat.category_id
            WHERE cat.category_name = :category;
        """
        df_comp = run_query(query_comp, params={"category": selected_cat})

    col_a, col_b = st.columns([1, 2])
    with col_a:
        st.metric("Tournaments Listed", f"{len(df_comp):,}")
        type_dist = df_comp['type'].value_counts().reset_index()
        type_dist.columns = ['Type', 'Count']
        fig_pie = px.pie(type_dist, names='Type', values='Count', title="Singles vs. Doubles
Distribution")
        st.plotly_chart(fig_pie, use_container_width=True)

    with col_b:
        st.dataframe(df_comp, use_container_width=True, height=450)
```

```
# ------------------------------------------------------------
# 3. Venue Infrastructure
# ------------------------------------------------------------
elif menu == "Venue Infrastructure":
    st.title("Global Venue & Stadium Network")

    sub_tab1, sub_tab2 = st.tabs(["Complexes Breakdown", "Geographic Concentration"])

    with sub_tab1:
        st.subheader("Major Multi-Court Complexes")
        df_complex = run_query("""
            SELECT c.complex_name, COUNT(v.venue_id) AS venue_count
            FROM Complexes c
            JOIN Venues v ON c.complex_id = v.complex_id
            GROUP BY c.complex_name
            HAVING COUNT(v.venue_id) > 1
            ORDER BY venue_count DESC
            LIMIT 15;
        """)
        fig_comp = px.bar(df_complex, x="complex_name", y="venue_count",
                          labels={"complex_name": "Complex Name", "venue_count": "Court / Venue
 Count"},
                          color="venue_count", color_continuous_scale="Blues")
        st.plotly_chart(fig_comp, use_container_width=True)

    with sub_tab2:
        st.subheader("Venues by Country")
        df_ctry = run_query("""
            SELECT country_name, COUNT(venue_id) AS total_venues
            FROM Venues
            GROUP BY country_name
            ORDER BY total_venues DESC;
        """)
        st.dataframe(df_ctry, use_container_width=True, height=400)

# ------------------------------------------------------------
# 4. Competitor Rankings
# ------------------------------------------------------------
elif menu == "Competitor Rankings":
    st.title("Player Leaderboards & Points")

    col_f1, col_f2 = st.columns(2)
    with col_f1:
        top_n = st.slider("Select Top N Competitors", min_value=5, max_value=50, value=10)
    with col_f2:
        sort_metric = st.selectbox("Sort By", ["Points", "Rank"])

    order_clause = "r.points DESC" if sort_metric == "Points" else "r.`rank` ASC"

    df_rankings = run_query(f"""
        SELECT c.name, c.country, r.`rank`, r.movement, r.points, r.competitions_played
        FROM Competitor_Rankings r
        JOIN Competitors c ON r.competitor_id = c.competitor_id
        ORDER BY {order_clause}
        LIMIT :limit;
    """, params={"limit": top_n})

    fig_rank = px.bar(df_rankings, x="name", y="points", color="country",
                      title=f"Top {top_n} Competitors by Points",
                      labels={"name": "Player Name", "points": "ATP/WTA Points", "country":
"Nation"})
    st.plotly_chart(fig_rank, use_container_width=True)

    st.dataframe(df_rankings, use_container_width=True)

# ------------------------------------------------------------
# 5. SQL Query Sandbox
# ------------------------------------------------------------
elif menu == "SQL Query Sandbox":
    st.title("Live SQL Query Executor")
    st.markdown("Execute verified queries against `tennis_analytics`.")
```

```
    default_query = "SELECT c.name, c.country, r.`rank`, r.points FROM Competitor_Rankings r
JOIN Competitors c ON r.competitor_id = c.competitor_id ORDER BY r.points DESC LIMIT 10;"
    user_sql = st.text_area("Input SQL Query", value=default_query, height=120)

    if st.button("Execute Query"):
        try:
            res_df = run_query(user_sql)
            st.success(f"Query returned {len(res_df)} rows.")
            st.dataframe(res_df, use_container_width=True)
        except Exception as err:
            st.error(f"SQL Execution Error: {err}")
```

```
Microsoft Windows [Version 10.0.26300.9457]
(c) Microsoft Corporation. All rights reserved.

C:\aFiles And Media\DataAnalytics\HexaSolutions\Hexa Assignments\Assignments To Submit 09-10-2026\End to End Game Analyt
ics>streamlit run "etega_stremlit_code.py"
2026-10-07 05:43:29.513 Uvicorn server started on :::8501

  You can now view your Streamlit app in your browser.

  Local URL: http://localhost:8501
  Network URL: http://10.115.195.149:8501

  Help agents write better Streamlit apps?
  Install the official Streamlit skills by running streamlit skills in your terminal.
```

Tennis Analytics

Navigate Modules

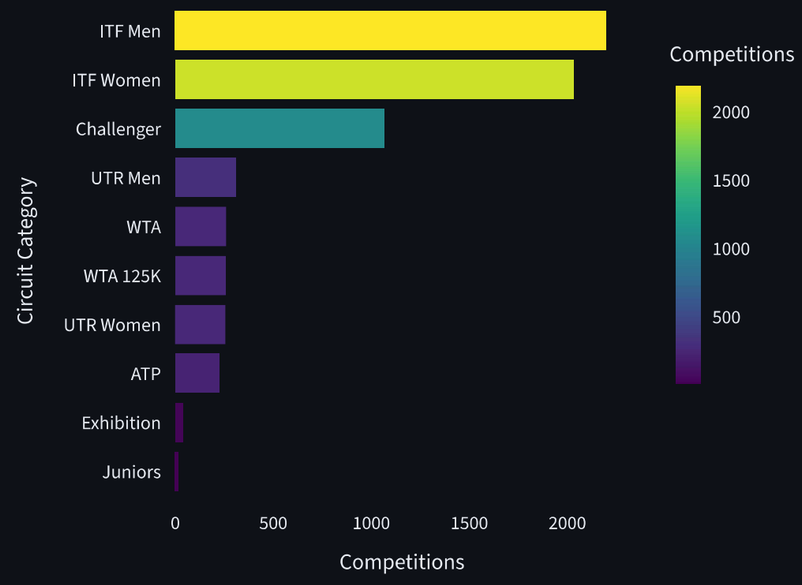
- (•) Executive Overview
- ( ) Competitions Explorer
- ( ) Venue Infrastructure
- ( ) Competitor Rankings
- ( ) SQL Query Sandbox

Tennis Ecosystem Global Overview

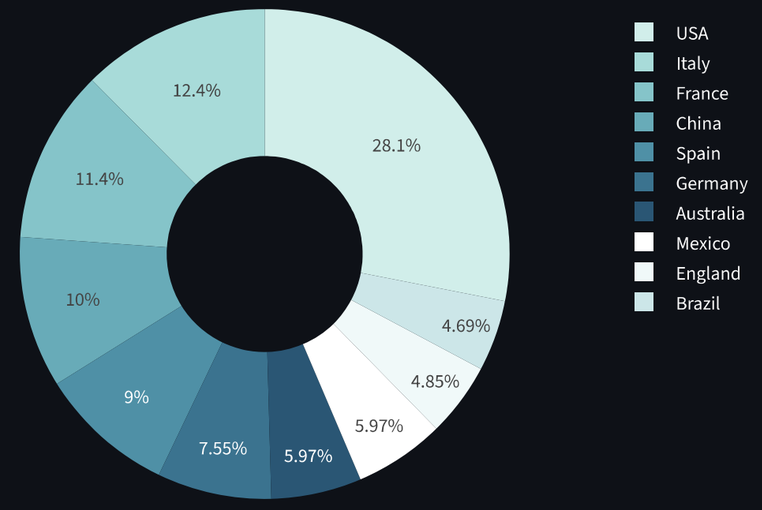
High-level KPI distribution across professional circuits, global venues, and active competitors.

Total Competitions	Total Venues	Tennis Complexes	Ranked Competitors
6,694	4,124	777	1,000

Tournament Volume by Category



Top 10 Host Countries by Venues



Tennis Analytics

Navigate Modules

- Executive Overview
- Competitions Explorer
- Venue Infrastructure
- Competitor Rankings
- SQL Query Sandbox

Competitions & Circuits Analysis

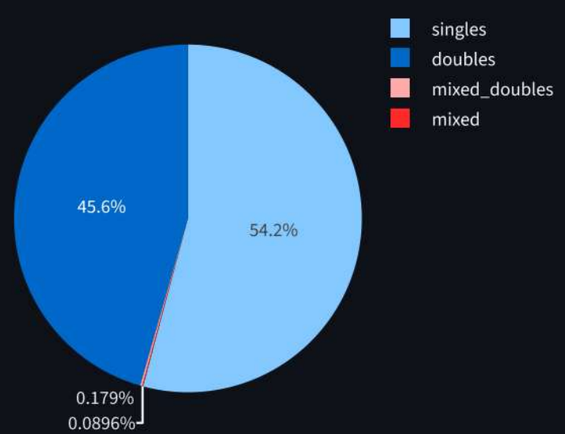
Filter by Category

All

Tournaments Listed

6,694

Singles vs. Doubles Distribution



	competition_id	competition_name	category_name	type	gender
0	sr:competition:10025	ITF Men Stara Zagora, Bulgaria Men Singles	ITF Men	singles	men
1	sr:competition:10027	ITF Men Stara Zagora, Bulgaria Men Doubles	ITF Men	doubles	men
2	sr:competition:10031	ITF Men Sibiu, Romania Men Singles	ITF Men	singles	men
3	sr:competition:10033	ITF Men Sibiu, Romania Men Doubles	ITF Men	doubles	men
4	sr:competition:10037	ITF Men Busto Arsizio, Italy Men Singles	ITF Men	singles	men
5	sr:competition:10039	ITF Men Busto Arsizio, Italy Men Doubles	ITF Men	doubles	men
6	sr:competition:10043	ITF Men Sabac, Serbia Men Singles	ITF Men	singles	men
7	sr:competition:10045	ITF Men Sabac, Serbia Men Doubles	ITF Men	doubles	men
8	sr:competition:10055	ITF Men Seefeld, Austria Men Singles	ITF Men	singles	men
9	sr:competition:10057	ITF Men Seefeld, Austria Men Doubles	ITF Men	doubles	men
10	sr:competition:10073	ATP Challenger Portoroz, Slovenia Men Single	Challenger	singles	men
11	sr:competition:10075	ATP Challenger Portoroz, Slovenia Men Doubl	Challenger	doubles	men

Tennis Analytics

Navigate Modules

- Executive Overview
- Competitions Explorer
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Global Venue & Stadium Network

Complexes Breakdown | Geographic Concentration

Major Multi-Court Complexes



Tennis Analytics

Navigate Modules

- Executive Overview
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Global Venue & Stadium Network

Complexes Breakdown | Geographic Concentration

Venues by Country

	country_name	total_venues
0	USA	677
1	Italy	300
2	France	276
3	China	241
4	Spain	217
5	Germany	182
6	Australia	144
7	Mexico	144
8	England	117
9	Brazil	113

Tennis Analytics

Navigate Modules

- Executive Overview
- Competitions Explorer
- Venue Infrastructure
- Competitor Rankings
- SQL Query Sandbox

Player Leaderboards & Points

Select Top N Competitors

10

Sort By

Points

Top 10 Competitors by Points



	name	country	rank	movement	points	competitions_played
0	Sinner, Jannik	Italy	1	0	11000	16
1	Rybakina, Elena	Kazakhstan	1	0	9901	21
2	Zverev, Alexander	Germany	2	0	9720	20
3	Sabalenka, Aryna	Neutral	2	0	7810	20
4	Pegula, Jessica	USA	3	0	7265	19
5	Gauff, Coco	USA	4	0	7244	19
6	Andreeva, Mirra	Neutral	5	0	5841	21
7	Noskova, Linda	Czechia	6	0	5328	20
8	Alcaraz, Carlos	Spain	3	0	5260	17
9	Svitolina, Elina	Ukraine	7	0	4809	17

Tennis Analytics

Navigate Modules

- Executive Overview
- Competitions Explorer
- Venue Infrastructure
- Competitor Rankings
- SQL Query Sandbox

Player Leaderboards & Points

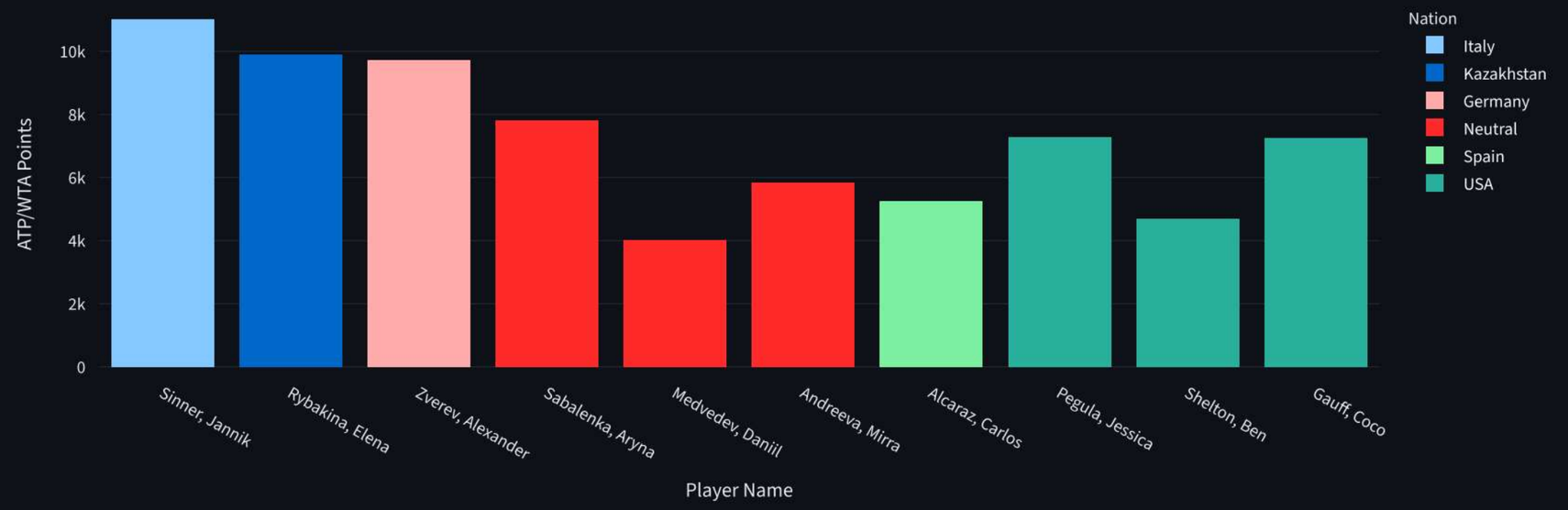
Select Top N Competitors

10

Sort By

Rank

Top 10 Competitors by Points



	name	country	rank	movement	points	competitions_played
0	Sinner, Jannik	Italy	1	0	11000	16
1	Rybakina, Elena	Kazakhstan	1	0	9901	21
2	Zverev, Alexander	Germany	2	0	9720	20
3	Sabalenka, Aryna	Neutral	2	0	7810	20
4	Alcaraz, Carlos	Spain	3	0	5260	17
5	Pegula, Jessica	USA	3	0	7265	19
6	Shelton, Ben	USA	4	0	4680	22
7	Gauff, Coco	USA	4	0	7244	19
8	Medvedev, Daniil	Neutral	5	1	4010	24
9	Andreeva, Mirra	Neutral	5	0	5841	21

Tennis Analytics

Navigate Modules

- ○ Executive Overview
- ○ Competitions Explorer
- ○ Venue Infrastructure
- ○ Competitor Rankings
- ◉ SQL Query Sandbox

Live SQL Query Executor

Execute verified queries against `tennis_analytics`.

Input SQL Query

```
SELECT c.name, c.country, r.`rank`, r.points FROM Competitor_Rankings r JOIN Competitors c ON r.competitor_id = c.competitor_id ORDER BY r.points DESC LIMIT 10;
```

Execute Query

Query returned 10 rows.

	name	country	rank	points
0	Sinner, Jannik	Italy	1	11000
1	Rybakina, Elena	Kazakhstan	1	9901
2	Zverev, Alexander	Germany	2	9720
3	Sabalenka, Aryna	Neutral	2	7810
4	Pegula, Jessica	USA	3	7265
5	Gauff, Coco	USA	4	7244
6	Andreeva, Mirra	Neutral	5	5841
7	Noskova, Linda	Czechia	6	5328
8	Alcaraz, Carlos	Spain	3	5260
9	Svitolina, Elina	Ukraine	7	4809

Assignment Insights & Business Conclusions

Based on the verified query outputs in your MySQL Workbench results:

1. Professional Tournament Ecosystem (Query 1.1 – 1.7)

- **Grassroots Pyramids Dominate Professional Play:** ITF Men (2,198 competitions) and ITF Women (2,032 competitions) comprise **63.2%** of all 6,694 recorded events. Elite tours (ATP: 225, WTA: 259) account for less than **7.3%** of the schedule, illustrating that tournament tennis is heavily supported by the lower-tier feeder circuits.
- **Balanced Formats:** Doubles events represent 3,051 competitions, closely mirroring singles play across challenger and senior circuits.
- **Hierarchy Structure:** Out of 6,694 competitions, 608 operate as top-level standalone tournaments without parents (parent_id IS NULL), while sub-events (singles and doubles draws under a common regional circuit) link together via relational keys.

2. Physical Infrastructure & Venue Footprint (Query 2.1 – 2.7)

- **Major Infrastructure Hubs:** The database covers 4,124 venues across 756 complexes worldwide. Multi-venue concentration is high: 583 complexes maintain multiple courts, headed by facilities like the **National Tennis Center (75 venues)**, Buenos Aires Lawn Tennis Club (29), and Melbourne Park (28).
- **Western Geographic Clustering:** The United States (677 venues), Italy (300), France (276), and China (241) account for the largest shares of host facilities, reflecting historical tournament investment and facility density.

3. Competitor Performance & Nationality Trends (Query 3.1 – 3.6)

- **Top-Tier Dominance:** The point distribution shows an elite concentration at the summit (Jannik Sinner leading at 11,000 points, Elena Rybakina at 9,901, and Alexander Zverev at 9,720 points).
- **Rank Stability:** Out of the 1,000 recorded ranking slots, 90 competitors experienced **0 rank movement**, indicating high placement stability at the upper tiers during the recording window.
- **Talent Pools:** Player representations are led by the USA (100 competitors), France (77), and Italy (66), aligning with the nations that maintain the highest density of physical tournament venues.